
University of Kansas

**Group 18 Minesweeper
Hours**

Actual Person Hours

1. WEEK 1

1.1 September 6th

Team Member	Task Completed	Hours Spent
Jake	Planning	1
John	Planning	1
Ahmed	Reviewed documentation requirements	1
Cameren	Planning	1
Karim	Planning	1
Zema		
Nate	planning	1
Pranav	planning	1

1.2 September 7th

Team Member	Task Completed	Hours Spent
Jake	Meeting and communicating with John on planning	3
John	Meeting, taking notes, and communicating with Jake on planning	3
Ahmed		
Cameren	Group meeting	1
Karim	Group Meeting	1
Zema	Group meeting	1
Nate	Group meeting	1
Pranav	Group meeting	1

1.3 September 8th

Team Member	Task Completed	Hours Spent
Jake	Meeting/Made Github	1.5
John	Meeting/drafted a template	1.5
Ahmed		
Cameren		
Karim	Meeting/Made GitHub	1
Zema		
Nate	Meeting	1
Pranav		

1.4 September 9th

Team Member	Task Completed	Hours Spent
Jake	Introduction & High Level Architecture and met with John on assigning tasks to TEAM.	2.5
John	Assumptions/constraints and met with Jake on assigning tasks	2.5
Ahmed		
Cameren		
Karim	Met with Zema to start planning on	2

	how to distribute the work and how we want to do it	
Zema		
Nate		
Pranav		

1.5 September 10th

Team Member	Task Completed	Hours Spent
Jake	Team meeting and Architecture work	1
John	Team meeting and architecture work.	1
Ahmed	Updated person-hours log	1
Cameren	Team meeting and architecture work	1
Karim	Team Meeting	1
Zema	Team meeting	1
Nate	Team meeting and architecture work	1
Pranav	Team meeting and architecture work	1

1.6 September 11th

Team Member	Task Completed	Hours Spent
Jake	-	-
John	-	-
Ahmed		
Cameren	Meeting with Pranav to review project requirements	1
Karim	Met with Zema to work on our portion of the project	2
Zema		
Nate		
Pranav	Meeting with cam to review project requirements	1

1.7 September 12th

Team Member	Task Completed	Hours Spent
Jake	-	-
John	-	-
Ahmed		
Cameren		
Karim	-	-
Zema		
Nate		
Pranav		

2. WEEK 2

2.1 September 13th

Team Member	Task Completed	Hours Spent
Jake	Reviewing implementation ideas for group / Organizational Planning	1
John	Reviewed implementation ideas for	1

	group meetings.	
Ahmed	Drafted data flow documentation	2
Cameren		
Karim		
Zema		
Nate		
Pranav	Working on architecture sections 5.1-5.3	2

2.2 September 14th

Team Member	Task Completed	Hours Spent
Jake	-	-
John	-	-
Ahmed		
Cameren	Meeting with Pranav regarding sprites architecture	1
Karim		
Zema		
Nate		
Pranav	Meeting with Cam regarding sprites architecture	1

2.3 September 15th

Team Member	Task Completed	Hours Spent
Jake	Started working on the design and worked on the architecture.	1.5
John	Started implementation design and worked on architecture.	1.5
Ahmed		
Cameren		
Karim		
Zema		
Nate		
Pranav		

2.4 September 16th

Team Member	Task Completed	Hours Spent
Jake	Meeting/ planning implementation	2
John	Meeting/planning implementation/taking meeting notes.	2
Ahmed	Checked person-hours entries and totals	1
Cameren	Meeting/ planning implementation w/ Pranav	3
Karim		
Zema	Meeting/drafted and reviewed online tutorials and structures for our implementation to follow	5
Nate	Draft flowchart	2
Pranav	Meeting/ planning implementation	3

	w/ Cam	
--	--------	--

2.5 September 17th

Team Member	Task Completed	Hours Spent
Jake	Worked on main.py and commenting / Team meeting	3
John	Worked on main.py and commenting, and attended team meeting.	3
Ahmed		
Cameren	Attended team meeting, worked on implementations	3
Karim		
Zema	Worked on implementing settings / team meeting	3
Nate	Work on flowchart	3
Pranav	Attended team meeting, worked on implementations	3

2.6 September 18th

Team Member	Task Completed	Hours Spent
Jake	Worked on main.py and updated comments.	4
John	Worked on main.py and updated commenting.	4
Ahmed	Reviewed main.py and sprites.py	1.5
Cameren	Reviewed sprites.py and looked over testing with Pranav	1
Karim		
Zema	Structured the files in the repository	1
Nate	Review Flow chart	1
Pranav	Reviewed sprites.py and looked over testing with cam	1

2.7 September 19th

Team Member	Task Completed	Hours Spent
Jake	Finished main.py, tested integration, completed, and updated comments. Coordinating Team Tasks / Organization.	5
John	Finished main.py, tested system integration, completed a README, and created an Actual Hours doc.	5
Ahmed	Proofread documentation and finalized hours log	1.5
Cameren	Finalized sprites.py	3
Karim		
Zema	Worked on my section of the architecture document and merged the branches to main.	4
Nate	Finalized Flow chart	3

Pranav	Finalized sprites.py with bug fixing	2
--------	--	---

3. Estimated Hours

Calculate Use Case Weight (UUCW)

Use case: "Configure Game" (Console input, mine count selection, validation)

- Complexity: Simple
- Weight: 5 points

Use case: "Uncover Tile" (Click, dynamic mine placement, recursive dig, win/loss check)

- Complexity: Average
- Weight: 10 points

Use case: "Toggle Flag" (Right click and update HUD counter)

- Complexity: Simple
- Weight: 5 points

Use case: "Restart Game" (End screen click, board reset, variable reinitialization)

- Complexity: Simple
- Weight: 5 points

Total UUCW = 5 + 10 + 5 + 5 = **25**

Calculate Actor Weight (UAW)

Actor: "Player" (Interacts via graphical mouse interface)

- Complexity: Complex
- Weight: 3 points

Actor: "System" (Pygame event loop and window management)

- Complexity: Simple
- Weight: 1 point

Total UAW = 3 + 1 = **4**

Calculate Unadjusted UCP (UUCP)

$$\text{UUCP} = \text{UUCW} + \text{UAW} = 25 + 4 = \mathbf{29}$$

Final Effort

Assume a 15% adjustment factor (1.15 multiplier) for recursive flood-fill and Pygame graphics handling:

$$\text{Adjusted UCP} = 29 \times 1.15 = \mathbf{33.35}$$

Assume a productivity rate of 3 person-hours per UCP due to the team's size and small scale of the project:

$$\text{Final Estimated Effort} = 33.35 \times 3 = \mathbf{100.05 \text{ person-hours}}$$